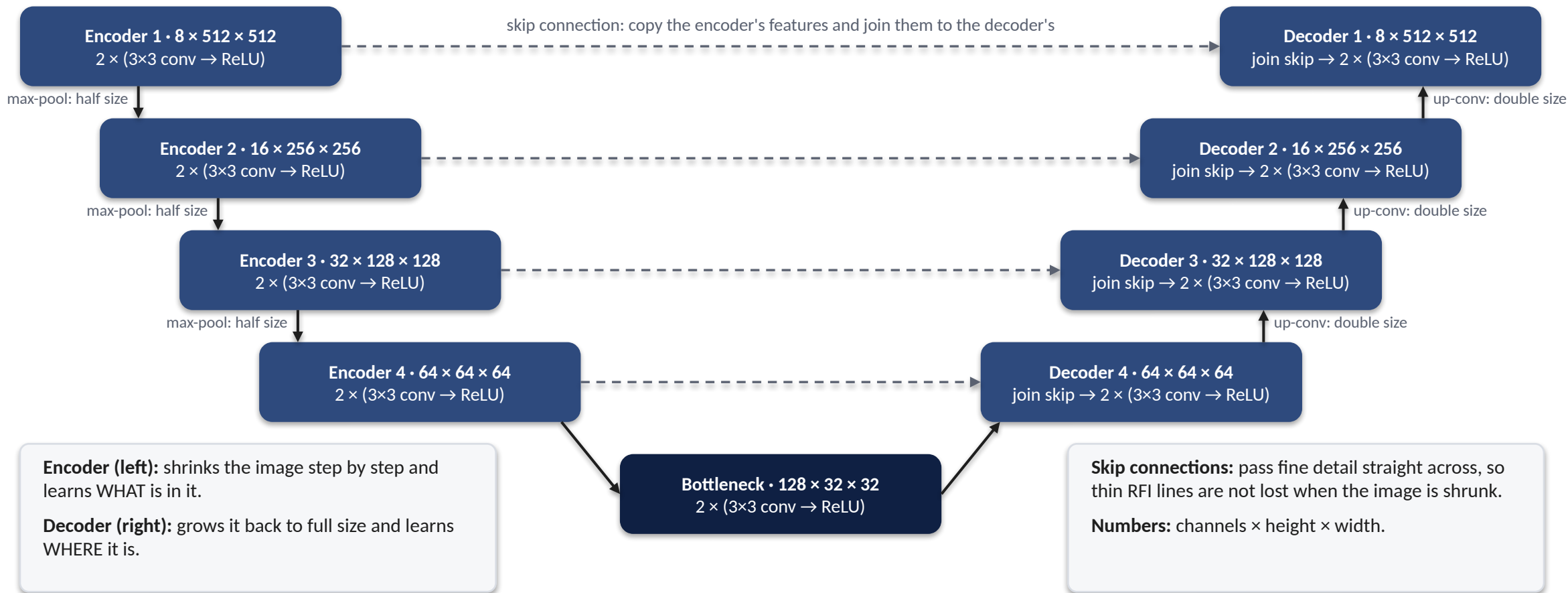


## ➤ Starting point: a basic U-Net

▼ Input: spectrogram,  $1 \times 512 \times 512$  (time  $\times$  frequency)

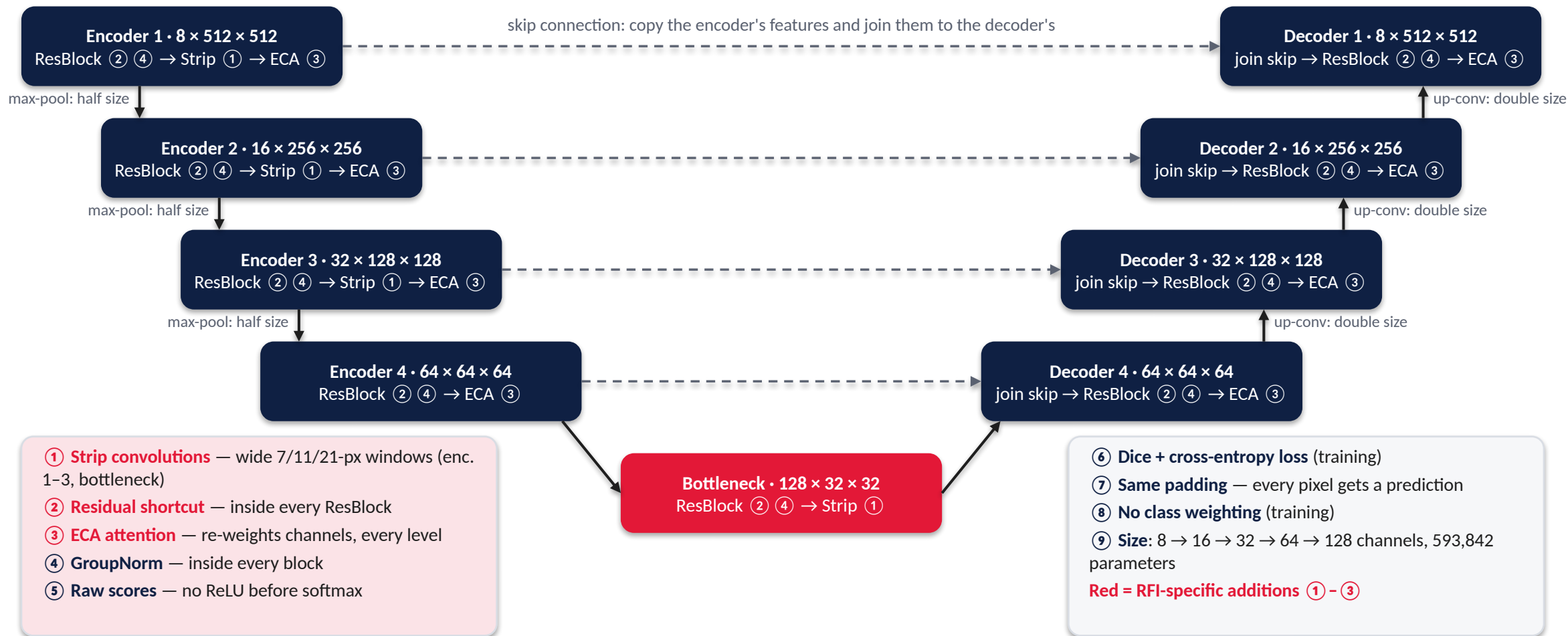
Output: an RFI probability for every pixel ▲



## ➤ Our model: the same U, with 9 changes

▼ Input: spectrogram,  $1 \times 512 \times 512$  — sizes never shrink ⑦

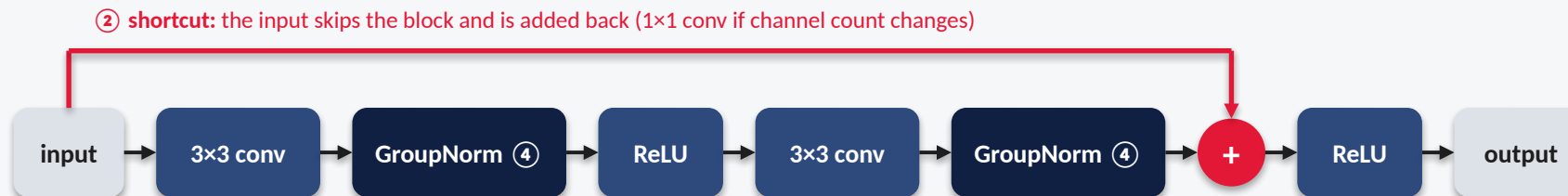
$1 \times 1$  conv  $\rightarrow$  2 raw scores  $\rightarrow$  softmax ⑤  $\rightarrow$  RFI probability ▲



## ➤ Inside one level, step by step

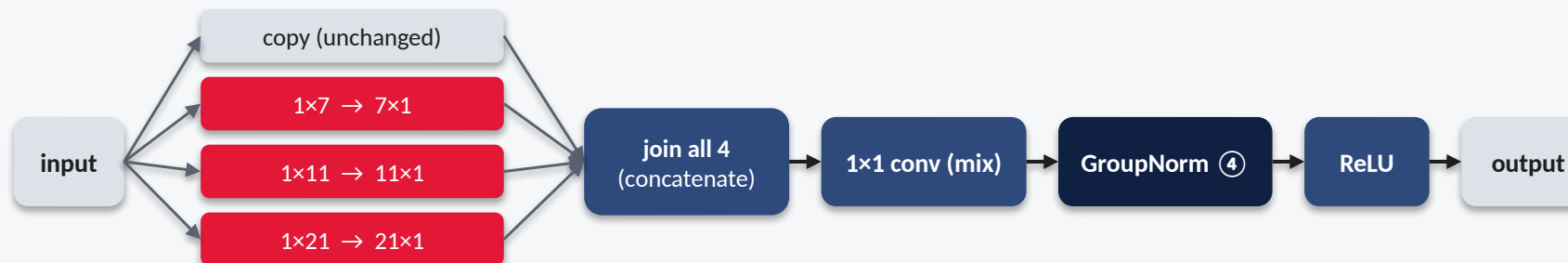
### ResBlock ② ④

Two 3×3 convolutions, each tidied by GroupNorm; the input is added back at the end.



### Strip module ①

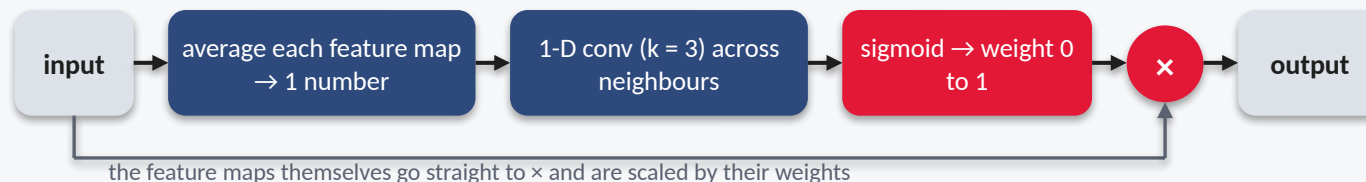
Looks at wider areas (7, 11, 21 px) cheaply: a thin filter along frequency, then one along time.



Each path = 1×K along frequency, then K×1 along time → together it covers a K×K square (measured)

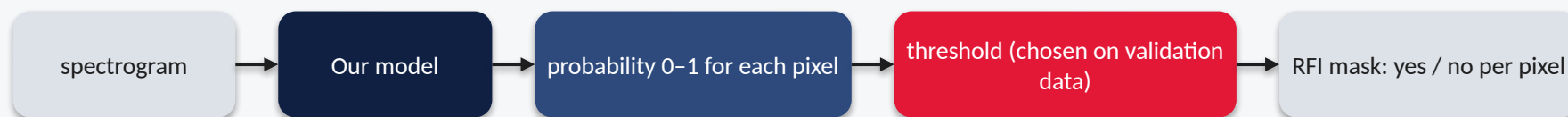
### ECA attention ③

Gives each feature map a weight between 0 and 1, then scales it.

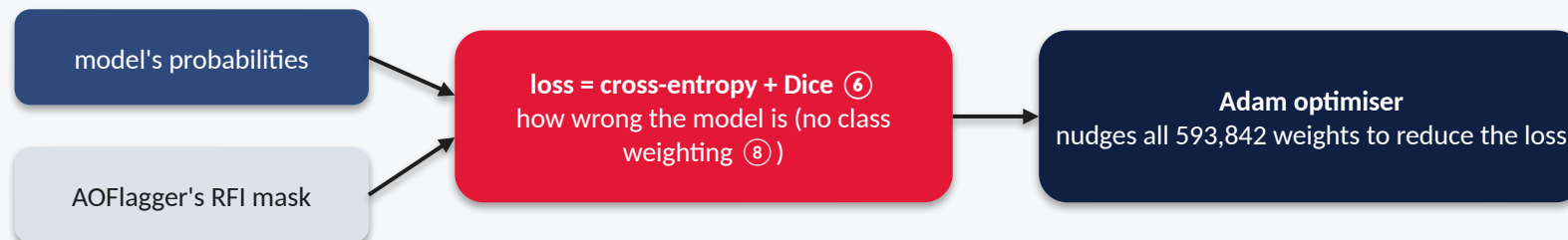


## ➤ From output to RFI mask, and how it learns

Using the model  
after training



Training  
repeated 28,000 times



**Which parts actually matter?** Removing ① ② ③ (strips, shortcuts, attention) changes F1 from 0.660 to 0.658 — not significant (3 runs each,  $p = 0.64$ ).

**So the model's strength comes from ④ - ⑨:** normalisation, raw output scores, the Dice loss, same padding, no class weighting, and its size and training setup.